Programming Fundamentals

Lab 3 Submission

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Lab 3 – Task 1

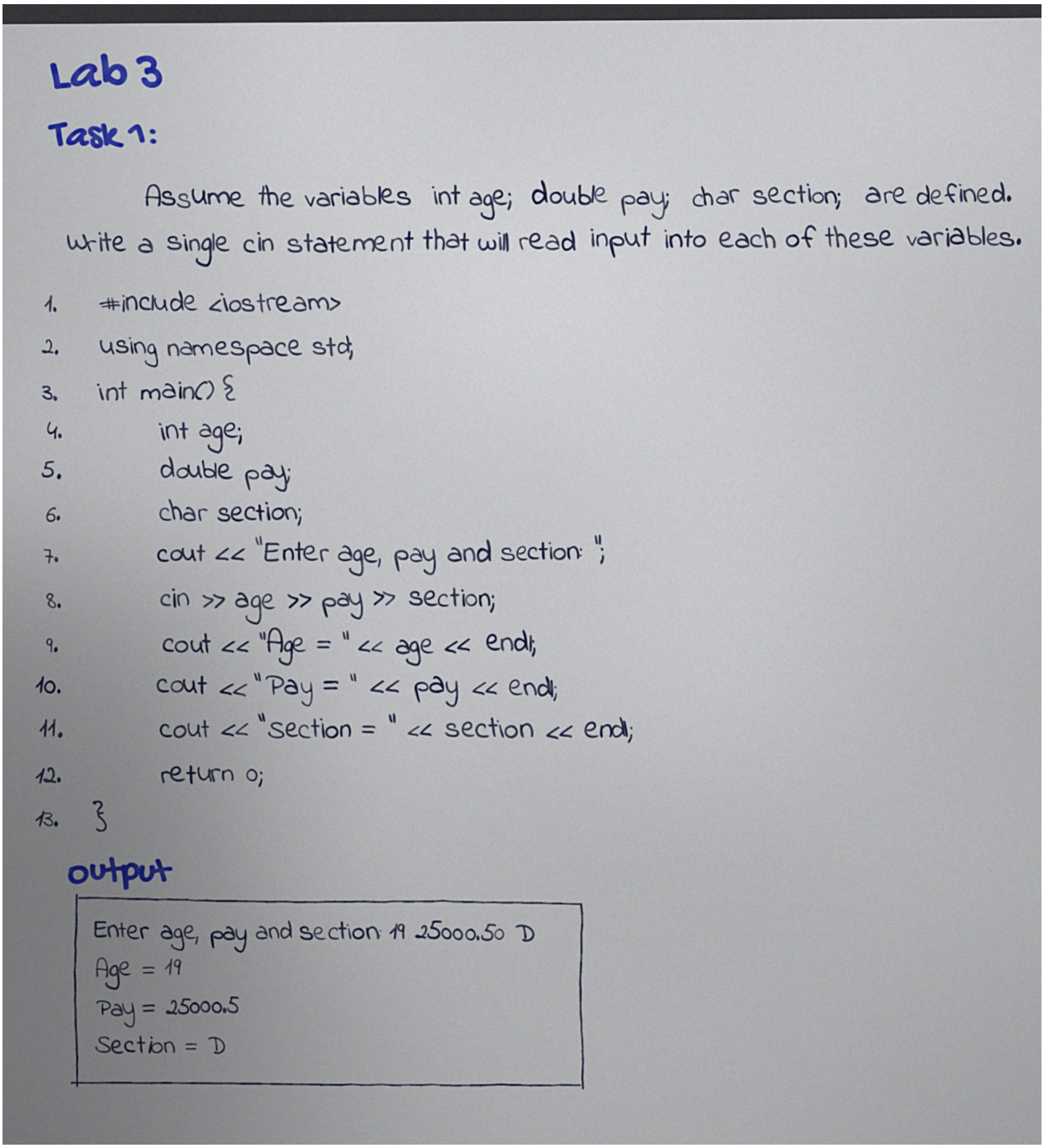
Question

Assume that the following variables are defined:

```
int age;
double pay;
char section;
```

Write a single cin statement that will read input into each of these variables.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int age;
    double pay;
    char section;
    cout << "Enter age, pay and section: ";
    cin >> age >> pay >> section;
    cout << "Age = " << age << endl;
    cout << "Pay = " << pay << endl;
    cout << "Section = " << section << endl;
    return 0;
}
```

Output

```
Enter age, pay and section: 19 25000.50 D
Age = 19
Pay = 25000.5
Section = D
```

Explanation

- The single statement is `cin >> age >> pay >> section;`.
- The values are typed with spaces between them, and `cin` stores each one in the next variable in order.
- `age` gets a whole number, `pay` gets a decimal number and `section` gets one character.

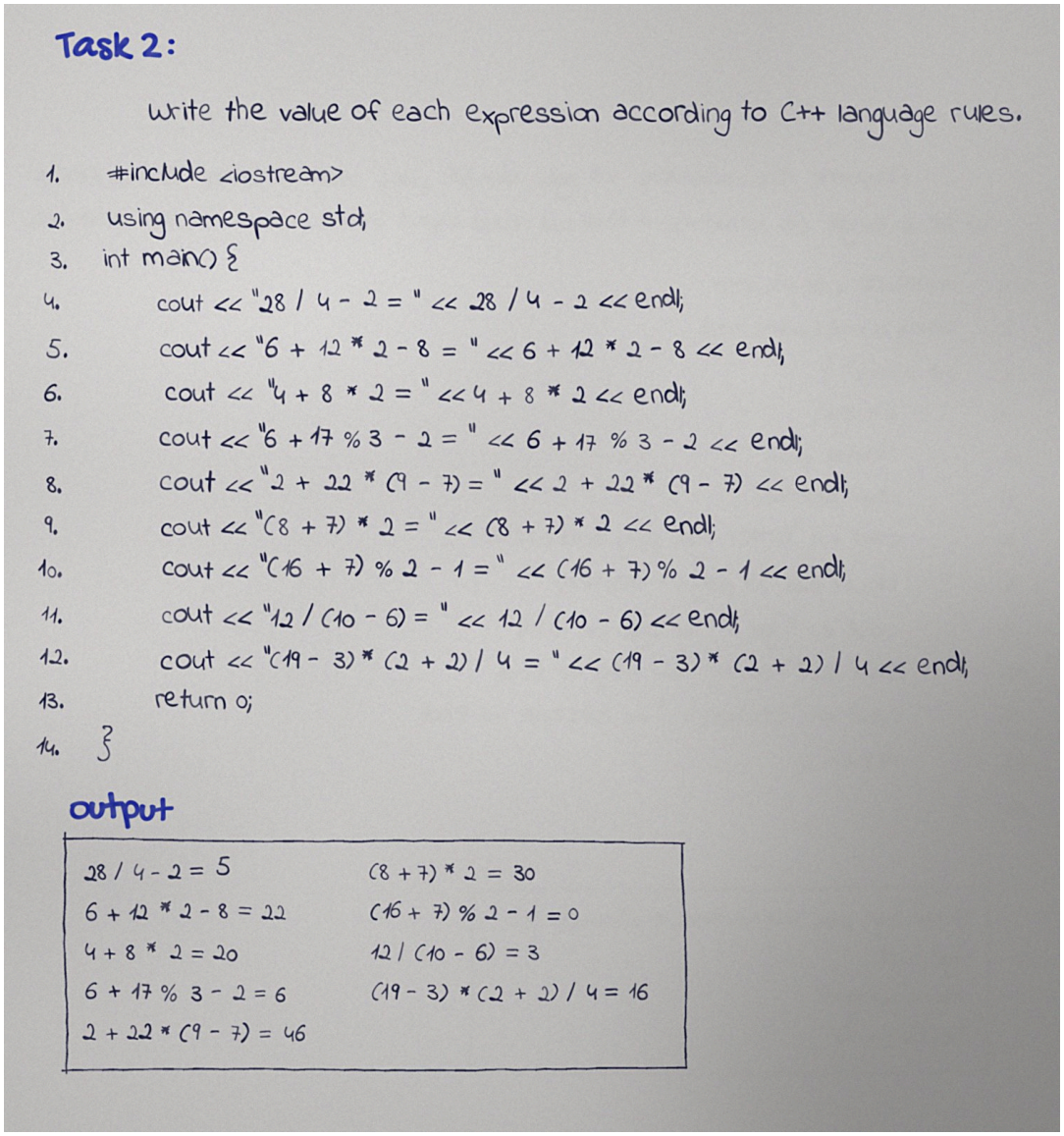
Lab 3 – Task 2

Question

Complete the following table by writing the value of each expression in the Value column according to C++ language rules.

```
28 / 4 - 2
6 + 12 * 2 - 8
4 + 8 * 2
6 + 17 % 3 - 2
2 + 22 * (9 - 7)
(8 + 7) * 2
(16 + 7) % 2 - 1
12 / (10 - 6)
(19 - 3) * (2 + 2) / 4
```

Handwritten Code



Task 2:

write the value of each expression according to C++ language rules.

```
1.  #include <iostream>
2.  using namespace std;
3.  int main() {
4.      cout << "28 / 4 - 2 = " << 28 / 4 - 2 << endl;
5.      cout << "6 + 12 * 2 - 8 = " << 6 + 12 * 2 - 8 << endl;
6.      cout << "4 + 8 * 2 = " << 4 + 8 * 2 << endl;
7.      cout << "6 + 17 % 3 - 2 = " << 6 + 17 % 3 - 2 << endl;
8.      cout << "2 + 22 * (9 - 7) = " << 2 + 22 * (9 - 7) << endl;
9.      cout << "(8 + 7) * 2 = " << (8 + 7) * 2 << endl;
10.     cout << "(16 + 7) % 2 - 1 = " << (16 + 7) % 2 - 1 << endl;
11.     cout << "12 / (10 - 6) = " << 12 / (10 - 6) << endl;
12.     cout << "(19 - 3) * (2 + 2) / 4 = " << (19 - 3) * (2 + 2) / 4 << endl;
13.     return 0;
14. }
```

output

```
28 / 4 - 2 = 5
6 + 12 * 2 - 8 = 22
4 + 8 * 2 = 20
6 + 17 % 3 - 2 = 6
2 + 22 * (9 - 7) = 46
(8 + 7) * 2 = 30
(16 + 7) % 2 - 1 = 0
12 / (10 - 6) = 3
(19 - 3) * (2 + 2) / 4 = 16
```

Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << "28 / 4 - 2 = " << 28 / 4 - 2 << endl;
    cout << "6 + 12 * 2 - 8 = " << 6 + 12 * 2 - 8 << endl;
    cout << "4 + 8 * 2 = " << 4 + 8 * 2 << endl;
    cout << "6 + 17 % 3 - 2 = " << 6 + 17 % 3 - 2 << endl;
    cout << "2 + 22 * (9 - 7) = " << 2 + 22 * (9 - 7) << endl;
    cout << "(8 + 7) * 2 = " << (8 + 7) * 2 << endl;
    cout << "(16 + 7) % 2 - 1 = " << (16 + 7) % 2 - 1 << endl;
    cout << "12 / (10 - 6) = " << 12 / (10 - 6) << endl;
    cout << "(19 - 3) * (2 + 2) / 4 = " << (19 - 3) * (2 + 2) / 4 << endl;
    return 0;
}
```

Output

```
28 / 4 - 2 = 5
6 + 12 * 2 - 8 = 22
4 + 8 * 2 = 20
6 + 17 % 3 - 2 = 6
2 + 22 * (9 - 7) = 46
(8 + 7) * 2 = 30
(16 + 7) % 2 - 1 = 0
12 / (10 - 6) = 3
(19 - 3) * (2 + 2) / 4 = 16
```

Explanation

- Brackets are solved first. Then `*`, `/` and `%` are done before `+` and `-`, going from left to right.
- For example, in `6 + 17 % 3 - 2` the remainder 17 % 3 = 2 is found first, so the answer is 6 + 2 - 2 = 6.
- All the numbers are integers, so `/` gives a whole number, e.g. 28 / 4 = 7.

Lab 3 – Task 3

Question

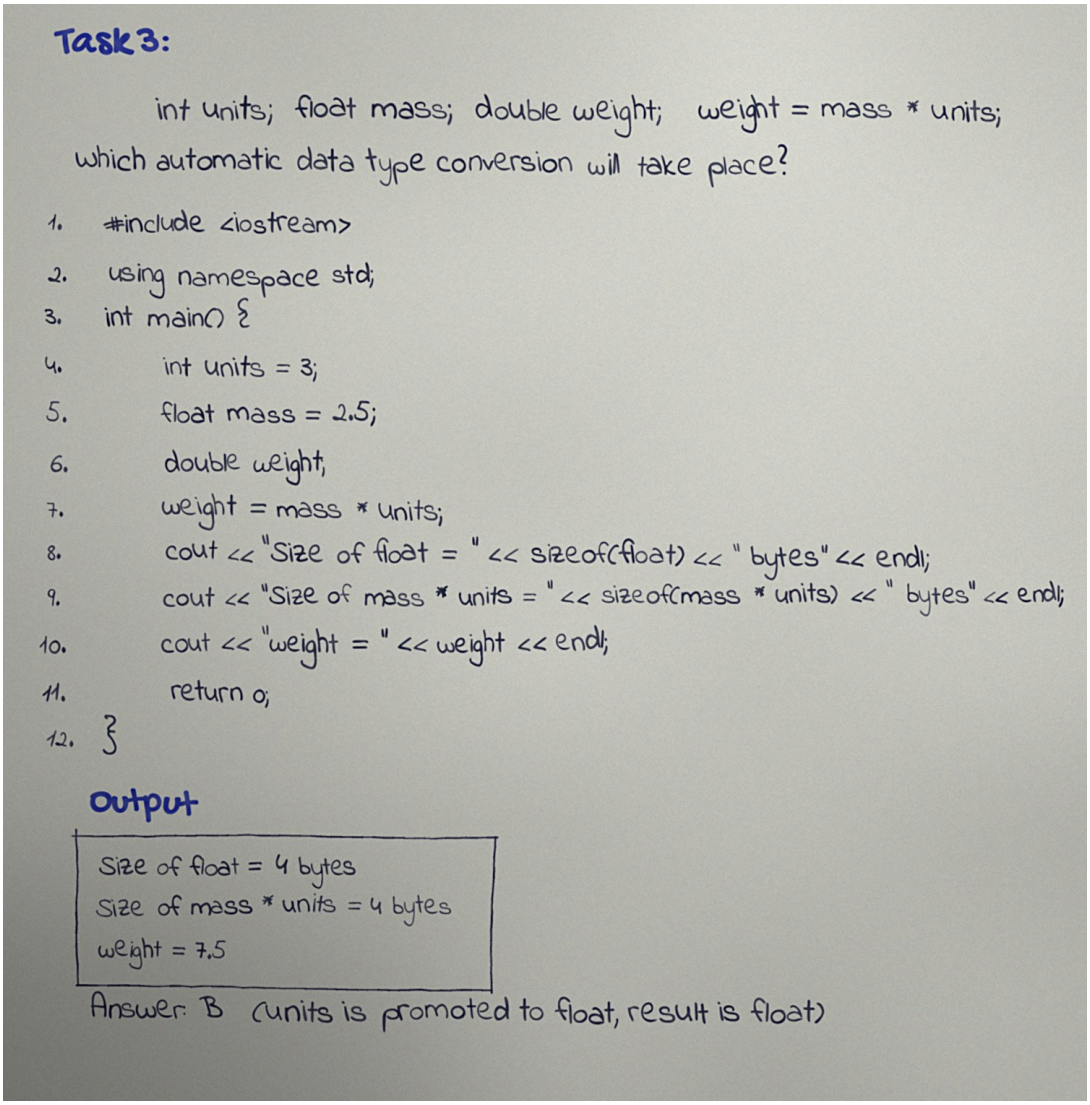
Assume a program has the following variable definitions:

```
int units;
float mass;
double weight;
weight = mass * units;
```

Which automatic data type conversion will take place?

```
A. mass is demoted to an int, units remains an int, and the result of
   mass * units is an int.
B. units is promoted to a float, mass remains a float, and the result of
   mass * units is a float.
C. units is promoted to a float, mass remains a float, and the result of
   mass * units is a double.
```

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int units = 3;
    float mass = 2.5;
    double weight;
    weight = mass * units;
    cout << "Size of float = " << sizeof(float) << " bytes" << endl;
    cout << "Size of mass * units = " << sizeof(mass * units) << " bytes" << endl;
    cout << "weight = " << weight << endl;
    return 0;
}
```

Output

```
Size of float = 4 bytes
Size of mass * units = 4 bytes
weight = 7.5
```

Explanation

- Answer: B.
- `units` is an int and `mass` is a float. The lower type (int) is promoted to the higher type (float), so the result of `mass * units` is a float.
- The program shows that `mass * units` has the same size as a float (4 bytes). The float result is only changed to double when it is stored in `weight`.

Lab 3 – Task 4

Question

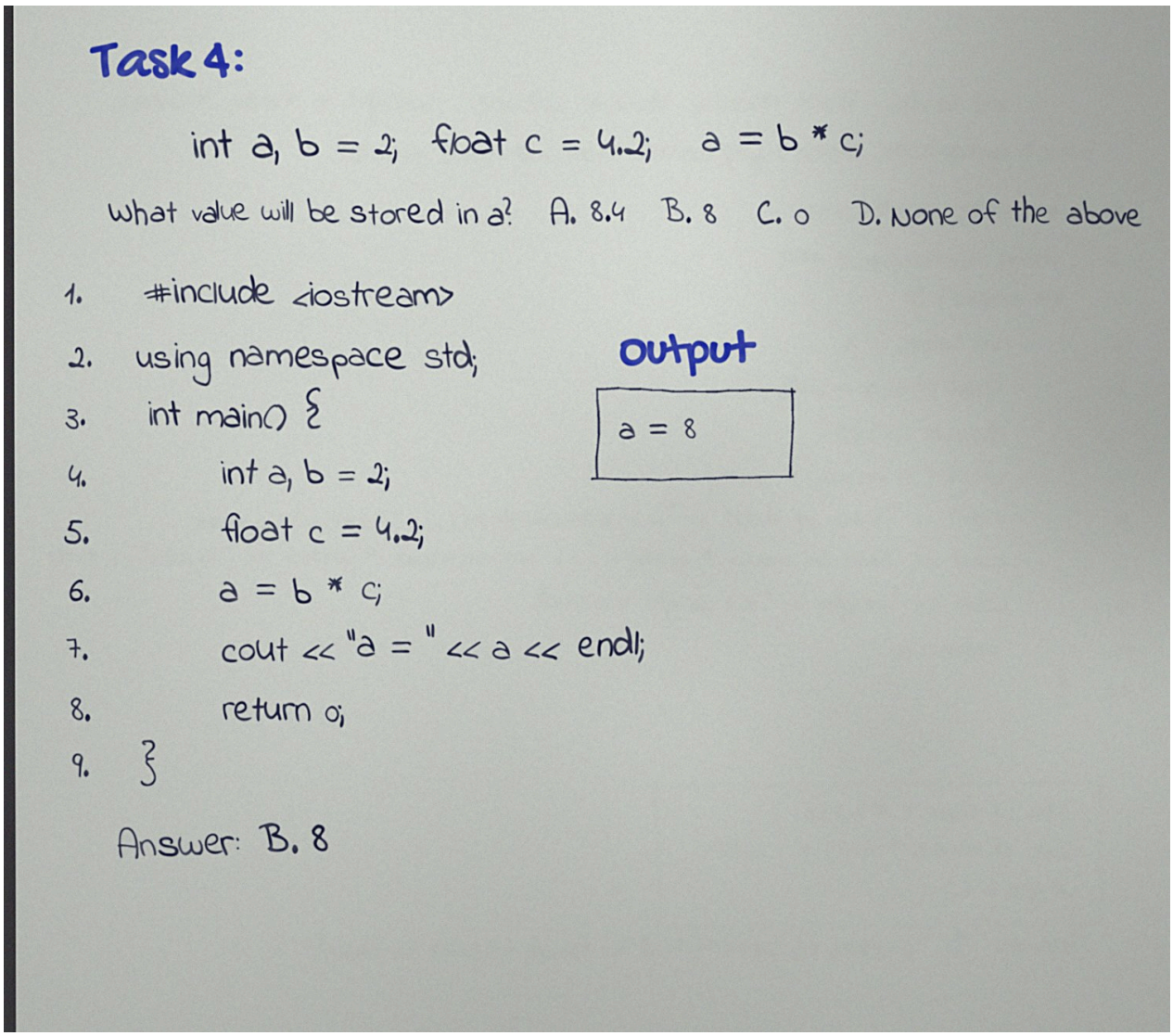
Assume a program has the following variable definitions:

```
int a, b = 2;     float c = 4.2;
```

and the following statement: a = b * c; What value will be stored in a?

```
A. 8.4      B. 8      C. 0      D. None of the above
```

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int a, b = 2;
    float c = 4.2;
    a = b * c;
    cout << "a = " << a << endl;
    return 0;
}
```

Output

```
a = 8
```

Explanation

- Answer: B. 8
- `b * c` is 2 * 4.2 = 8.4, which is a float value.
- `a` is an int, so when 8.4 is stored in it the decimal part is cut off and only 8 is stored.

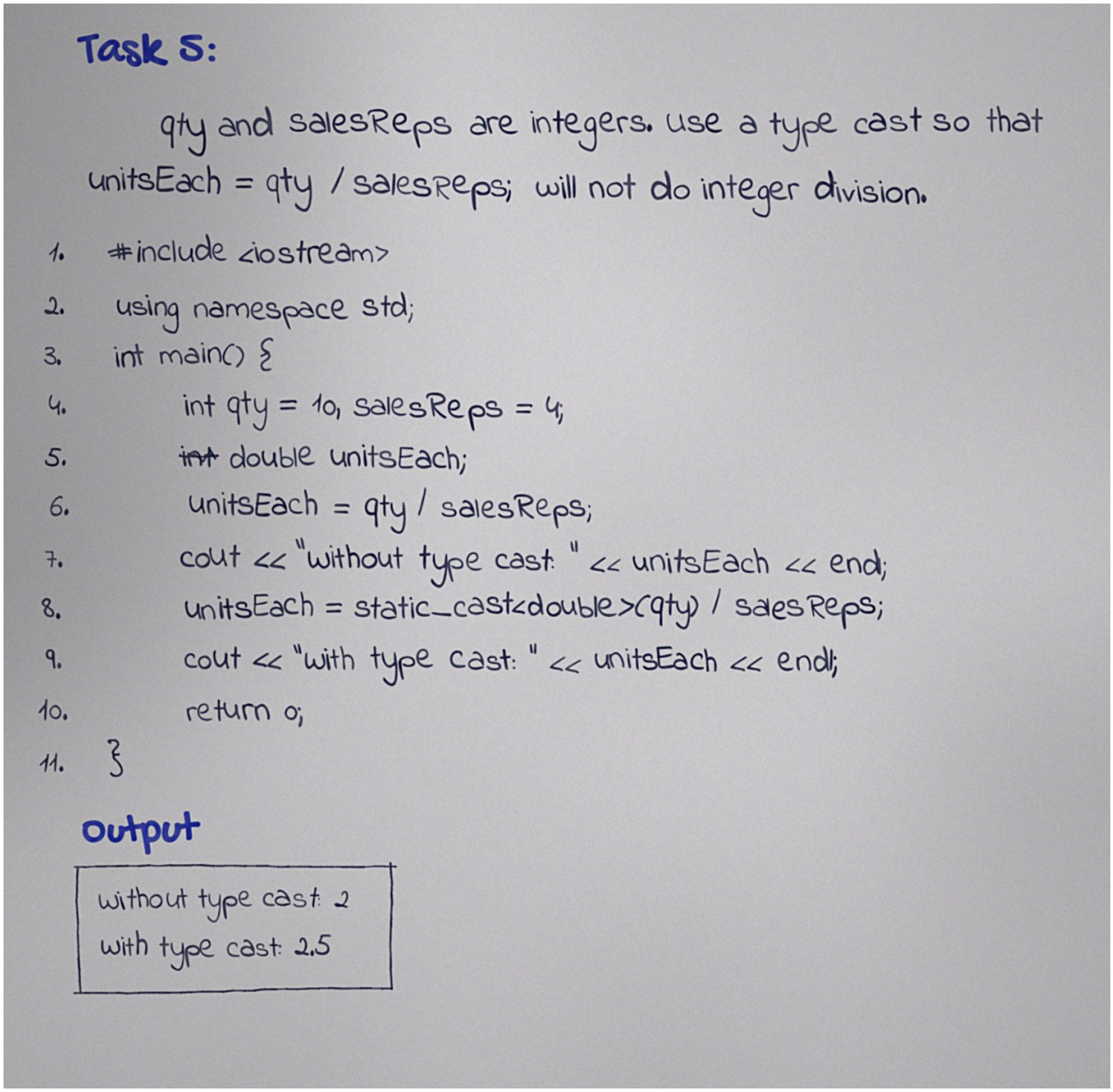
Lab 3 – Task 5

Question

Assume that qty and salesReps are both integers. Use a type cast expression to rewrite the following statement so it will no longer perform integer division.

```
unitsEach = qty / salesReps;
```

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int qty = 10, salesReps = 4;
    double unitsEach;
    unitsEach = qty / salesReps;
    cout << "Without type cast: " << unitsEach << endl;
    unitsEach = static_cast<double>(qty) / salesReps;
    cout << "With type cast: " << unitsEach << endl;
    return 0;
}
```

Output

```
Without type cast: 2
With type cast: 2.5
```

Explanation

- Answer: `unitsEach = static_cast<double>(qty) / salesReps;`
- Without the cast both values are int, so 10 / 4 gives 2 and the .5 is lost.
- `static_cast<double>(qty)` changes qty to a double first, so the division is done with decimals and gives 2.5.

Lab 3 – Task 6

Question

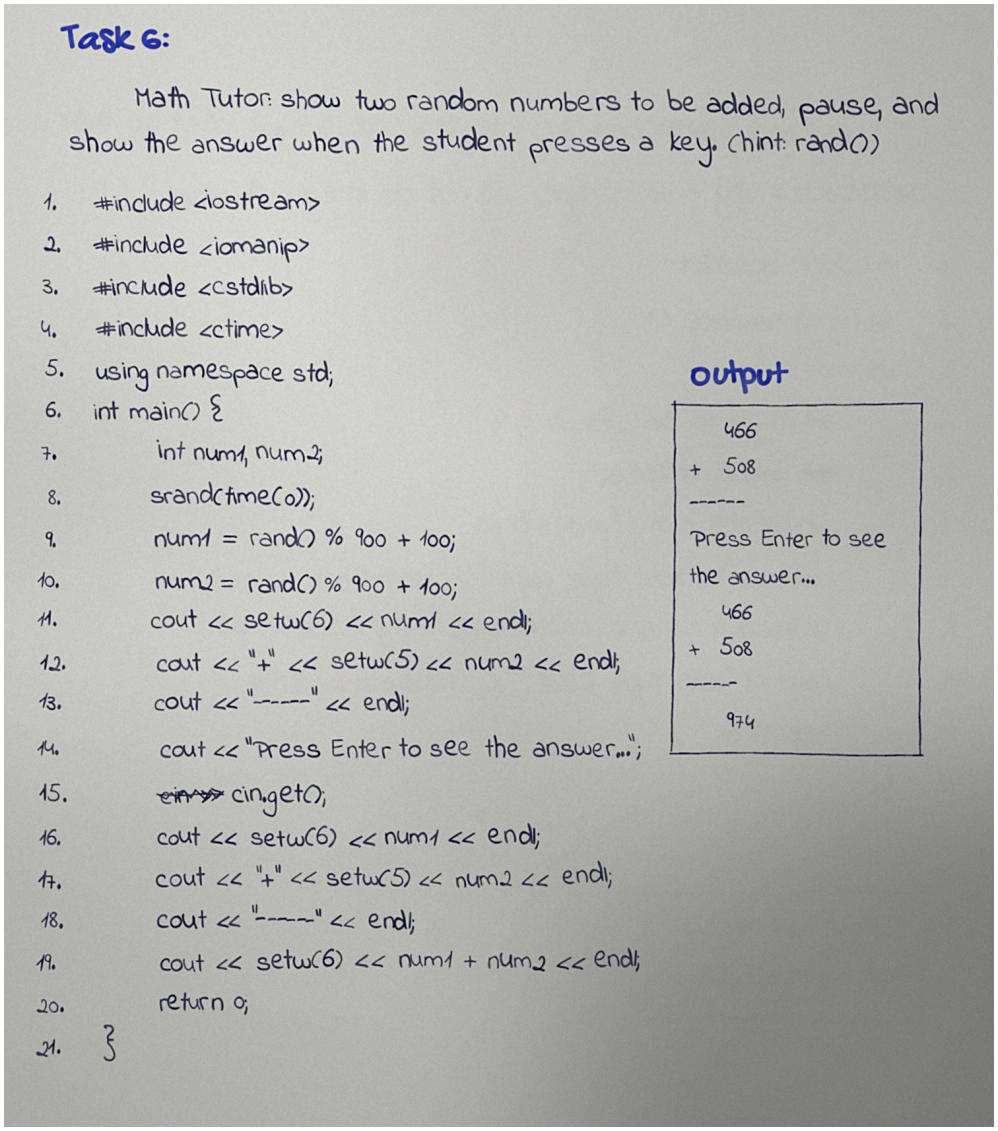
Math Tutor (hint: rand()): Write a program that can be used as a math tutor for a young student. The program should display two random numbers to be added, such as

```
   247
+  129
-----
```

The program should then pause while the student works on the problem. When the student is ready to check the answer, he or she can press a key and the program will display the correct solution:

```
   247
+  129
-----
   376
```

Handwritten Code



Task 6:

Math Tutor: show two random numbers to be added, pause, and show the answer when the student presses a key. (hint: rand())

```
1.  #include <iostream>
2.  #include <iomanip>
3.  #include <cstdlib>
4.  #include <ctime>
5.  using namespace std;
6.  int main() {
7.      int num1, num2;
8.      srand(time(0));
9.      num1 = rand() % 900 + 100;
10.     num2 = rand() % 900 + 100;
11.     cout << setw(6) << num1 << endl;
12.     cout << "+" << setw(5) << num2 << endl;
13.     cout << "-----" << endl;
14.     cout << "Press Enter to see the answer...";
15.     cin.get();
16.     cout << setw(6) << num1 << endl;
17.     cout << "+" << setw(5) << num2 << endl;
18.     cout << "-----" << endl;
19.     cout << setw(6) << num1 + num2 << endl;
20.     return 0;
21. }
```

output

```
   466
+  508
------
Press Enter to see
the answer...
   466
+  508
-----
   974
```

Code Written on Phone / Desktop

```cpp
#include <iostream>
#include <iomanip>
#include <cstdlib>
#include <ctime>
using namespace std;
int main() {
    int num1, num2;
    srand(time(0));
    num1 = rand() % 900 + 100;
    num2 = rand() % 900 + 100;
    cout << setw(6) << num1 << endl;
    cout << "+" << setw(5) << num2 << endl;
    cout << "------" << endl;
    cout << "Press Enter to see the answer...";
    cin.get();
    cout << setw(6) << num1 << endl;
    cout << "+" << setw(5) << num2 << endl;
    cout << "------" << endl;
    cout << setw(6) << num1 + num2 << endl;
    return 0;
}
```

Output

```
   466
+  508
------
Press Enter to see the answer...
   466
+  508
------
   974
```

Explanation

- `rand() % 900 + 100` gives a random 3-digit number (100 to 999). `rand()` comes from `<cstdlib>`.
- `srand(time(0))` uses the time as a seed, so the numbers change on every run.
- `setw()` lines the numbers up, and `cin.get()` waits for Enter before the answer is shown.

Lab 3 – Home Task 1

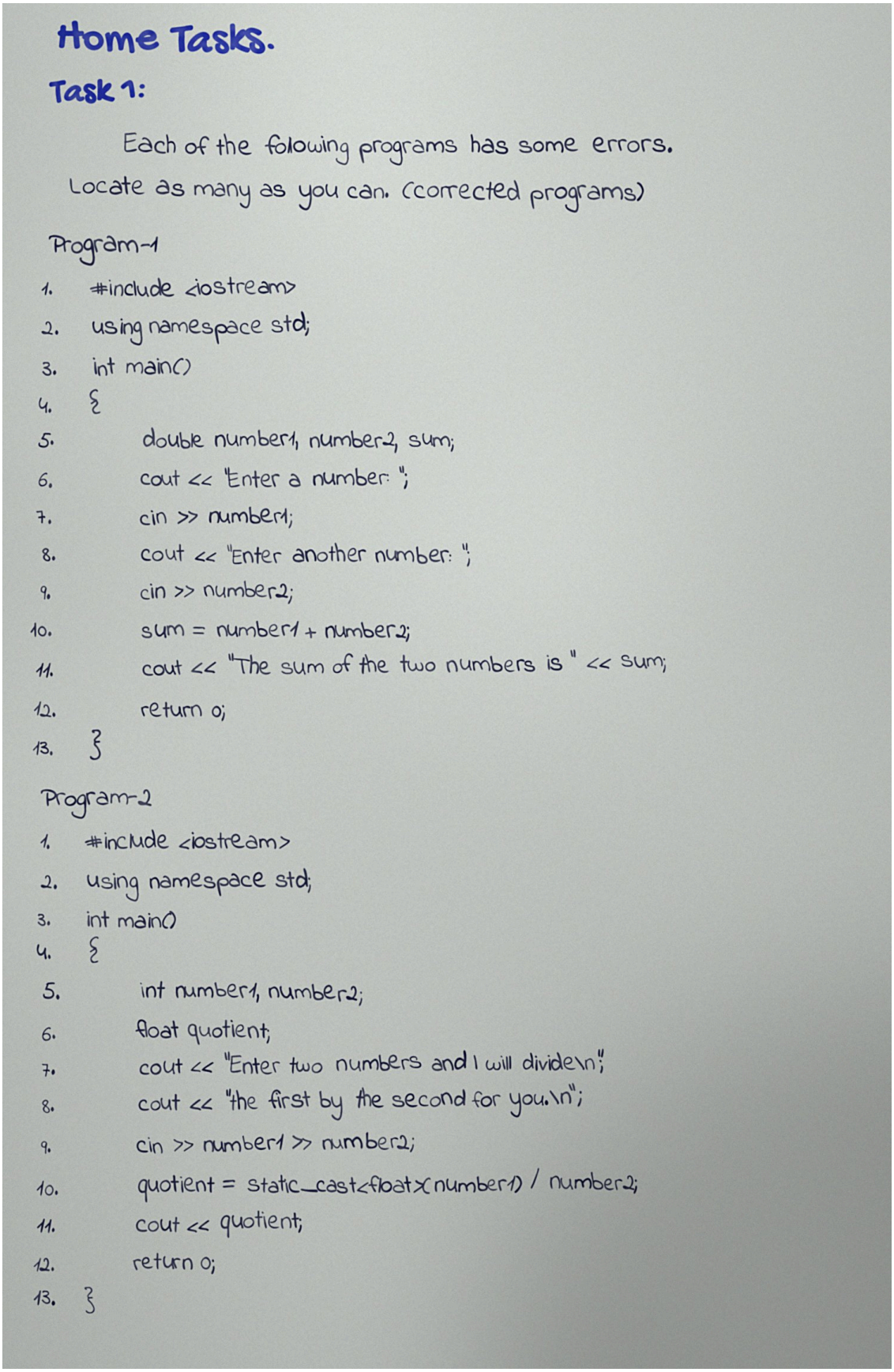
Question

Each of the following programs has some errors. Locate as many as you can.

```
Program-1
using namespace std;
void main ()
{
    double number1, number2, sum;
    cout << "Enter a number: ";
    Cin << number1;
    cout << "Enter another number: ";
    cin << number2;
    number1 + number2 = sum;
    cout "The sum of the two numbers is " << sum
}

Program-2
#include <iostream>
using namespace std;
void main()
{
    int number1, number2;
    float quotient;
    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";
    cin >> number1, number2;
    quotient = float<static_cast>(number1) / number2;
    cout << quotient
}
```

Handwritten Code



Home Tasks.

Task 1:

Each of the following programs has some errors. Locate as many as you can. (corrected programs)

Program-1

```
1.  #include <iostream>
2.  using namespace std;
3.  int main()
4.  {
5.      double number1, number2, sum;
6.      cout << "Enter a number: ";
7.      cin >> number1;
8.      cout << "Enter another number: ";
9.      cin >> number2;
10.     sum = number1 + number2;
11.     cout << "The sum of the two numbers is " << sum;
12.     return 0;
13. }
```

Program-2

```
1.  #include <iostream>
2.  using namespace std;
3.  int main()
4.  {
5.      int number1, number2;
6.      float quotient;
7.      cout << "Enter two numbers and I will divide\n";
8.      cout << "the first by the second for you.\n";
9.      cin >> number1 >> number2;
10.     quotient = static_cast<float>(number1) / number2;
11.     cout << quotient;
12.     return 0;
13. }
```

Code Written on Phone / Desktop (Program-1 corrected)

```cpp
#include <iostream>
using namespace std;
int main()
{
    double number1, number2, sum;
    cout << "Enter a number: ";
    cin >> number1;
    cout << "Enter another number: ";
    cin >> number2;
    sum = number1 + number2;
    cout << "The sum of the two numbers is " << sum;
    return 0;
}
```

Output (Program-1 corrected)

```
Enter a number: 12.5
Enter another number: 7.5
The sum of the two numbers is 20
```

Code Written on Phone / Desktop (Program-2 corrected)

```cpp
#include <iostream>
using namespace std;
int main()
{
    int number1, number2;
    float quotient;
    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";
    cin >> number1 >> number2;
    quotient = static_cast<float>(number1) / number2;
    cout << quotient;
    return 0;
}
```

Output (Program-2 corrected)

```
Enter two numbers and I will divide
the first by the second for you.
7 2
3.5
```

Errors Found

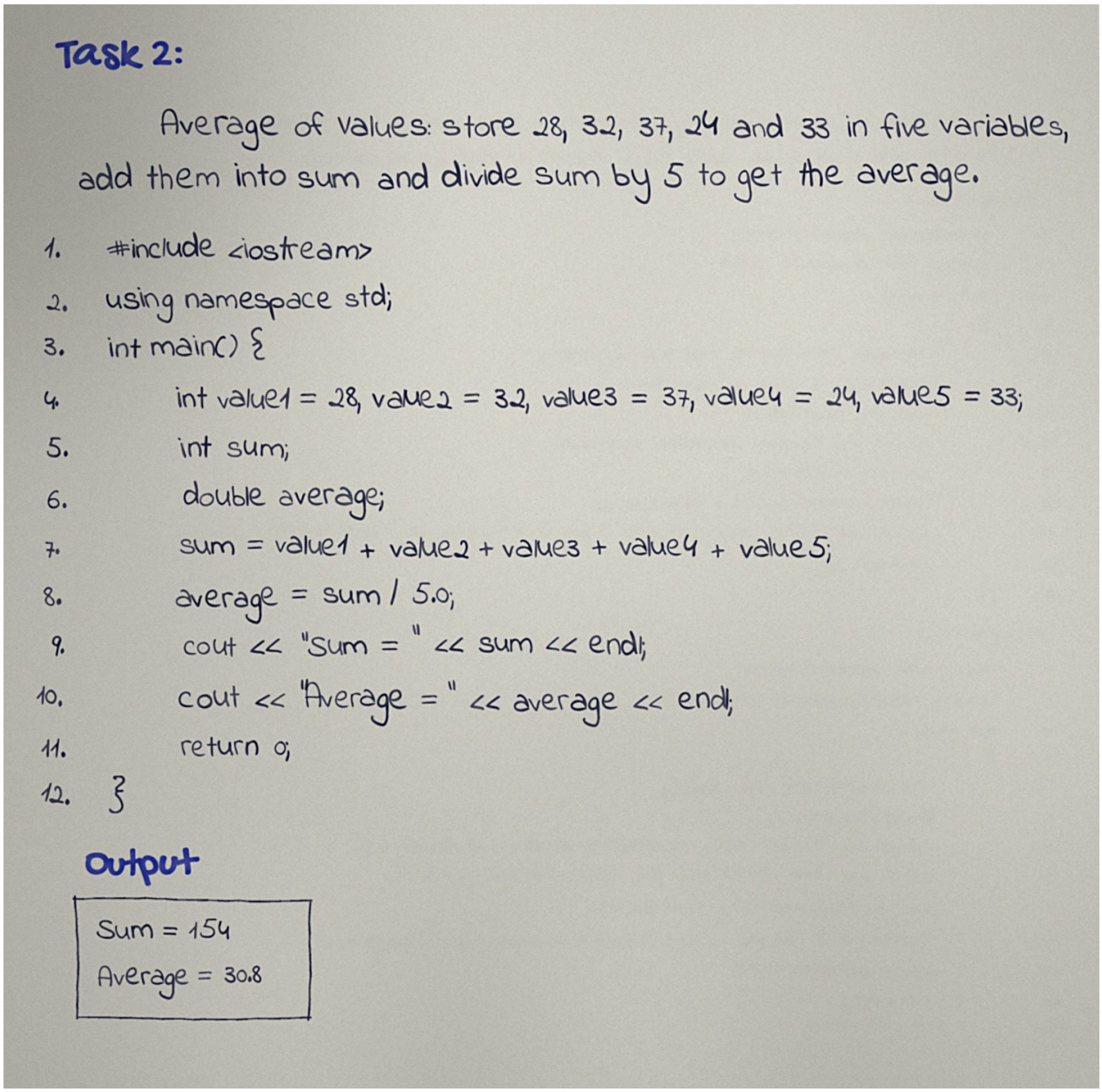
- Program-1: `#include <iostream>` is missing, so `cout` and `cin` are not known.
- Program-1: `void main()` should be `int main()` with `return 0;` at the end.
- Program-1: `Cin` should be `cin` (C++ is case sensitive), and `cin <<` should be `cin >>` in both input lines.
- Program-1: `number1 + number2 = sum;` is backwards. It should be `sum = number1 + number2;`
- Program-1: the last `cout` is missing `<<` before the string and a semicolon at the end.
- Program-2: `void main()` should be `int main()` with `return 0;`
- Program-2: `cin >> number1, number2;` only reads number1. It should be `cin >> number1 >> number2;`
- Program-2: `float<static_cast>(number1)` is written the wrong way round. It should be `static_cast<float>(number1)`.
- Program-2: `cout << quotient` is missing a semicolon.

Lab 3 – Home Task 2

Question

Average of Values: To get the average of a series of values, you add the values up and then divide the sum by the number of values. Write a program that stores the following values in five different variables: 28, 32, 37, 24, and 33. The program should first calculate the sum of these five variables and store the result in a separate variable named sum. Then, the program should divide the sum variable by 5 to get the average. Display the average on the screen.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int value1 = 28, value2 = 32, value3 = 37, value4 = 24, value5 = 33;
    int sum;
    double average;
    sum = value1 + value2 + value3 + value4 + value5;
    average = sum / 5.0;
    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;
    return 0;
}
```

Output

```
Sum = 154
Average = 30.8
```

Explanation

- The five values are stored in five separate variables and added into `sum`, which is 154.
- `sum / 5.0` is used instead of `sum / 5`. With 5 the answer would be integer division (30), but 5.0 makes it a decimal division, so the average is 30.8.

END OF ASSIGNMENT

```cpp
return 0;   // program finished successfully
```

Thank you for reading!